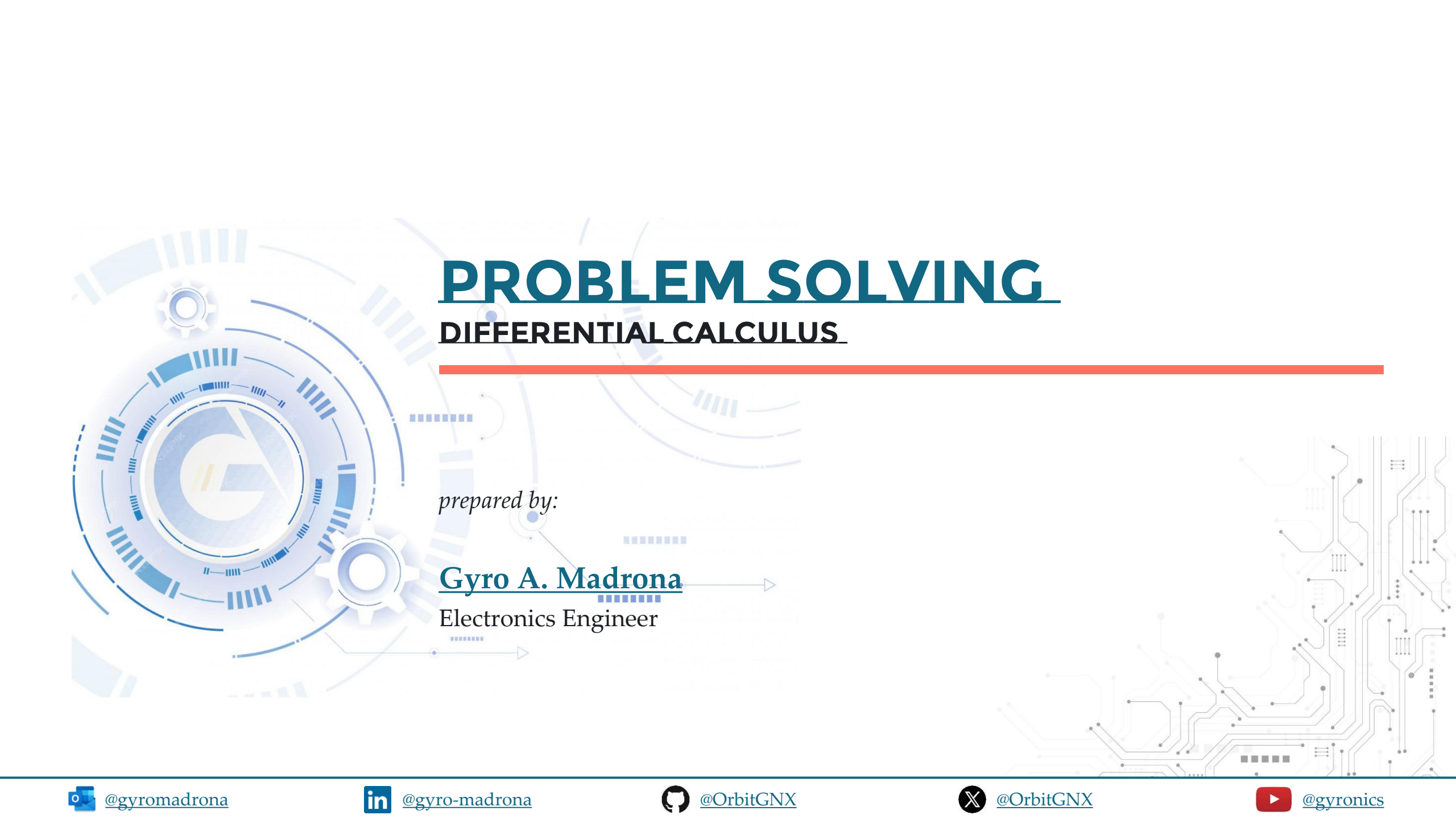




PROBLEM SOLVING

DIFFERENTIAL CALCULUS

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TOPIC OUTLINE

Problem Solving

- Free-falling Body
- Rate of Volume
- Rate of Profit



PROBLEM EXERCISE



EXERCISE

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground.

- What is the velocity of the ball after 5 seconds?
- How fast is the ball traveling when it hits the ground?

The distance fallen by a freely falling body is defined by the equation $\underline{s(t)} = 4.9t^2$.

↑ distance
as a function of time



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The CN Tower in Toronto was the tallest free-standing building in the world for 32 years.



EXERCISE

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground.

- What is the **velocity** of the ball after **5 seconds**?
- How fast is the ball traveling when it hits the ground?

The distance fallen by a freely falling body is defined by the equation $s(t) = 4.9t^2$.

Solution

Rate of change of distance

$$\frac{d}{dt} s(t) = \frac{d}{dt} (4.9t^2)$$

$$\frac{d}{dt} s(t) = 9.8t$$

The velocity after 5 sec

$$\text{let } t = 5$$

$$\frac{d}{dt} s(5) = 9.8(5)$$

$$\boxed{\frac{d}{dt} s(5) = 49 \text{ m/s}}$$

ans



EXERCISE

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground.

$s(t)$ ↗

- What is the velocity of the ball after 5 seconds?
- How fast is the ball traveling when it hits the ground?

The distance fallen by a freely falling body is defined by the equation $s(t) = 4.9t^2$.

Solution

The time it takes to fall from 450 m

$$\text{Let } s(t) = 450$$

$$\frac{450}{4.9} = \frac{4.9t^2}{4.9}$$

$$t = \sqrt{\frac{450}{4.9}}$$

$$\underline{t = 9.58 \text{ s}}$$



EXERCISE

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground.

$s(t)$ ↗

- What is the velocity of the ball after 5 seconds?
- How fast is the ball traveling when it hits the ground? $t = 9.58\text{s}$

The distance fallen by a freely falling body is defined by the equation $s(t) = 4.9t^2$.

Solution

The velocity after 9.58 sec

let $t = 9.58$

$$\frac{dy}{dt}s(t) = 9.8t$$

$$\frac{dy}{dt}s(9.58) = 9.8(9.58)$$

$$\frac{dy}{dt}s(9.58) = 93.88 \text{ m/s}$$

ans



EXERCISE

A conical pile of soil is being formed as soil pours from a spout, with the height of the cone always being twice the radius of the base. Determine the rate at which the volume of the cone is changing with respect to its height when the height is 4 feet.

$$\frac{dV}{dh} = ?$$

The volume of the cone is defined by the equation

$$V = \frac{1}{3}\pi r^2 h.$$

Solution

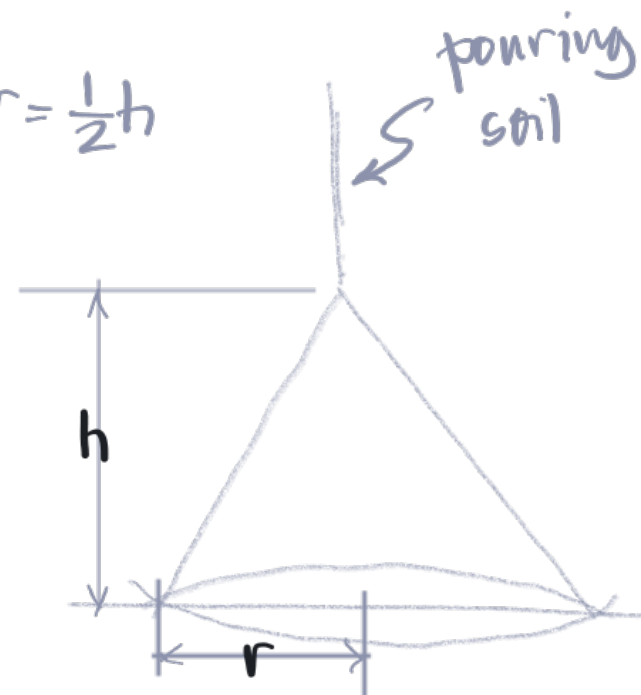
$$\text{Let } h = 2r \text{ then, } r = \frac{1}{2}h$$

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi \left(\frac{1}{2}h\right)^2 h$$

$$V = \frac{1}{3}\pi \left(\frac{1}{4}h^2\right) h$$

$$\underline{V(h) = \frac{1}{12}\pi h^3}$$



EXERCISE

A conical pile of soil is being formed as soil pours from a spout, with the height of the cone always being twice the radius of the base. Determine the rate at which the volume of the cone is changing with respect to its height when the height is 4 feet.

$$\frac{dV}{dh} = ?$$

The volume of the cone is defined by the equation

$$V = \frac{1}{3}\pi r^2 h.$$

Solution

Rate of change of volume w/ respect to height

$$V(h) = \frac{1}{12}\pi h^3$$

$$\frac{d}{dh} V(h) = \frac{d}{dh} \left(\frac{1}{12}\pi h^3 \right)$$

$$\frac{d}{dh} V(h) = \frac{1}{12}\pi \frac{d}{dh} (h^3)$$

$$\frac{d}{dh} V(h) = \frac{1}{12}\pi (3h^2)$$

$$\frac{d}{dh} V(h) = \frac{1}{4}\pi h^2$$



EXERCISE

A conical pile of soil is being formed as soil pours from a spout, with the height of the cone always being twice the radius of the base. Determine the rate at which the volume of the cone is changing with respect to its height when the height is 4 feet.

$$\frac{dV}{dh} = ?$$

The volume of the cone is defined by the equation

$$V = \frac{1}{3}\pi r^2 h.$$

Solution

$$\text{let } h = 4$$

$$\frac{dV}{dh} V(h) = \frac{1}{4} \pi h^2$$

$$\frac{dV}{dh} V(4) = \frac{1}{4} \pi (4)^2$$

$$\frac{dV}{dh} V(4) = \frac{1}{4} \pi (16)$$

$$\frac{dV}{dh} V(4) = 4\pi$$

$$\frac{dV}{dh} V(4) = 12.56 \text{ ft}^3/\text{s}$$

ans



EXERCISE

A growing e-commerce company is analyzing its financial performance over time. The company observes that its revenue $R(t)$ in dollars after some time t in months is modeled by the equation :

$$R(t) = 200t^2 + 500t + 2000$$

At the same time, the company's expenses grow as it hires more employees and scales its operation, modeled by:

$$E(t) = 300t + 1000$$

Determine the rate of change of the company's profit after 6 months.

Solution

profit is revenue less expenses

$$P(t) = R(t) - E(t)$$

$$P(t) = (200t^2 + 500t + 2000) \\ - (300t + 1000)$$

$$P(t) = 200t^2 + 200t + 1000$$



EXERCISE

A growing e-commerce company is analyzing its financial performance over time. The company observes that its revenue $R(t)$ in dollars after some time t in months is modeled by the equation :

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At the same time, the company's expenses grow as it hires more employees and scales its operation, modeled by:

$$E(t) = 300t + 1000$$

Determine the rate of change of the company's profit after 6 months.

Solution

Rate of change of profit w/ respect to time

$$\frac{d}{dt} P(t) = \frac{d}{dt} (200t^2 + 200t + 1000)$$

$$\frac{d}{dt} P(t) = 400t + 200$$

Let $t = 6$

$$\frac{d}{dt} P(6) = 400(6) + 200$$

$$\frac{d}{dt} P(6) = 2600 \text{ dollars/month}$$

ans



EXERCISE

A semiconductor manufacturing company tracks the output of wafers over time as it optimizes its production line. The number of wafers produced $W(t)$ after some time t in weeks is modeled by :

$$W(t) = 50t^2 + 200t + 1000$$

However, the defective wafers also increases due to wear and tear on the equipment is modeled by:

$$D(t) = 30t + 200$$

Determine the rate of change of **net usable wafers** after **4 weeks**.

Solution

Usable wafer is total production less defects

$$U(t) = W(t) - D(t)$$

$$U(t) = (50t^2 + 200t + 1000) - (30t + 200)$$

$$U(t) = 50t^2 + 170t + 800$$



EXERCISE

A semiconductor manufacturing company tracks the output of wafers over time as it optimizes its production line. The number of wafers produced $W(t)$ after some time t in weeks is modeled by :

$$W(t) = 50t^2 + 200t + 1000$$

However, the defective wafers also increases due to wear and tear on the equipment is modeled by:

$$D(t) = 30t + 200$$

Determine the rate of change of **net usable wafers** after **4 weeks**.

Solution

Rate of change of usable wafer over time

$$\frac{d}{dt} W(t) = \frac{d}{dt} (50t^2 + 170t + 800)$$

$$\frac{d}{dt} W(t) = 100t + 170$$

Let $t = 4$

$$\frac{d}{dt} W(4) = 100(4) + 170$$

$$\frac{d}{dt} W(4) = 570 \text{ wafers/week}$$

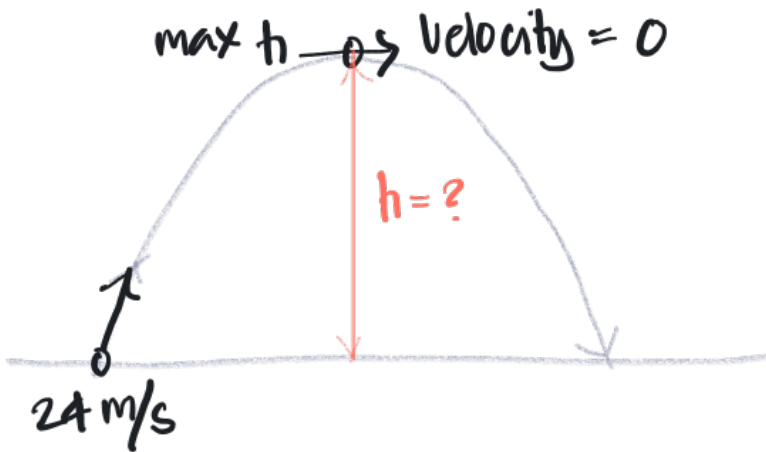
ans



EXERCISE

A rock thrown vertically upward from the surface of the earth at a velocity of 24 m/sec ^{initial velocity} reaches a height of $s = 24t - 4.9t^2$ meters in t seconds. How **high** would the **rock** go?

Hint: What happens to the velocity at maximum height?



Solution

Rate of change of distance over time

$$\frac{d}{dt} s(t) = \frac{d}{dt} (24t - 4.9t^2)$$

$$\frac{d}{dt} s(t) = 24 - 9.8t$$

The time it takes to reach max height

Let velocity = 0

$$0 = 24 - 9.8t$$

$$9.8t = 24$$

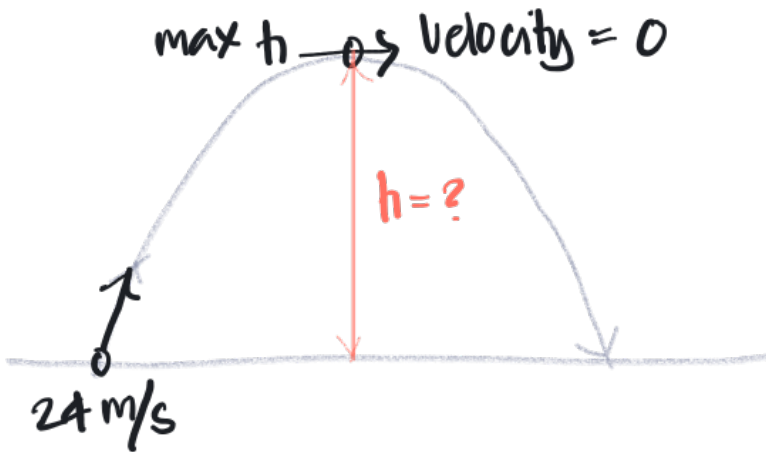
$$\underline{t = 2.45 \text{ s}}$$



EXERCISE

A rock thrown vertically upward from the surface of the earth at a velocity of 24 m/sec ^{initial velocity} reaches a height of $s = 24t - 4.9t^2$ meters in t seconds. How **high** would the **rock** go?

Hint: What happens to the velocity at maximum height?



Solution

Distance traveled after 2.45 s

$$\text{let } t = 2.45$$

$$s(t) = 24t - 4.9t^2$$

$$s(2.45) = 24(2.45) - 4.9(2.45)^2$$

$$s(2.45) = 29.39 \text{ m}$$

ans



LABORATORY

